AMATH 353 HW 7

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May 30, 2026

Question 1: An Easier Shock Solution

One setting where the method of characteristics fails is when the characteristics intersect and the solution becomes infinitely steep. This leads to the formation of *shock wave*, and we saw that the shock moves along a characteristic $(x_s(t), t)$ which can be determined from Rankine-Hugoniot condition

$$\frac{dx_s(t)}{dt} = \frac{\phi(u^+) - \phi(u^-)}{u^+ - u^-}.$$

Enforcing the Rankine-Hugoniot condition, determine the shock solution of the IVP

$$\begin{cases} u_t + uu_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}, \end{cases} \quad \text{where} \quad u_0(x) = \begin{cases} 2, & x < -1, \\ 1 - x, & |x| \leq 1. \\ 0, & x > 1. \end{cases}$$

Find the characteristic curve.

$$u_t + uu_x = 0$$
$$c(u) = u$$
$$\frac{dx}{dt} = u$$
$$u = u_0(x_0)$$
$$\frac{dx}{dt} = u$$
$$dx = udt$$
$$\int dx = \int u\, dt$$
$$x(t) = u_0(x_0)t + C$$

Solving for C

$$x(0) = x_0$$
$$x_0 = u_0(x_0)(0) + C$$
$$x_0 = C$$

$$x(t) = u_0(x_0)t + x_0$$

For the first region, $x_0 < -1,\ u_0(x_0) = 2$

$$x(t) = 2t + x_0$$

For the second region $x_0 > 0,\ u_0 = 0$

$$x(t) = (0)t + x_0 = X_0$$

For the third region where the break occurs $-1 \leq x \leq 1,\ u_0(x_0) = 1 - x_0$

$$\begin{aligned}
u &= 1 - x_0 \\
x(t) &= (1 - x_0)t + x_0 \\
x_0 &= \frac{x - t}{1 - t} \\
u(x,t) &= \frac{1 - x}{1 - t} \\
t_s &= 1
\end{aligned}$$

$$\begin{aligned}
\phi(u) &= \frac{u^2}{2} \\
u^- &= 2,\ u^+ = 0 \\
\frac{ds}{dt} &= \frac{0-2}{0-2} = 1 \\
x_s(t) &= t
\end{aligned}$$

Final piece wise solution:

$$\text{For } t \leq 1$$

$$u(x,t) = \begin{cases} 2,\ x < 2t - 1 \\ \frac{1-x}{1-t},\ 2t - 1 \leq x \leq 1 \\ 0,\ x > 1 \end{cases}$$

$$\text{For } t > 1$$

$$u(x,t) = \begin{cases} 2,\ x < t \\ 0,\ x > t \end{cases}$$

Question 2: A Harder Shock Solution

Enforcing the Rankine-Hugoniot condition, determine the shock solution of the IVP

$$\begin{cases} u_t + uu_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}, \end{cases} \quad \text{where} \quad u_0(x) = \begin{cases} 0, & x < 0, \\ x, & 0 \leq x \leq 1. \\ 0, & 1 < x. \end{cases}$$

For the first region, $x_0 < 0,\ u_0(x_0) = 0$

$$x(t) = (0)t + x_0$$
$$x = x_0$$
$$u(x,t) = 0$$

For the second region, $0 \leq x_0 \leq 1,\ u_0(x_0) = x_0$

$$x(t) = x_0 t + x_0 = x_0(t+1)$$
$$x_0 = \frac{x}{t+1}$$
$$u(x,t) = \frac{x}{t+1}$$

For the third region, $x_0 > 1,\ u_0(x_0) = 0$

$$x(t) = (0)t + x_0$$
$$x = x_0$$
$$u(x,t) = 0$$

Taking the derivative of the x_s with respect to t to find the shock speed between region 2 and 3:

$$t_s = 0,\ x_s(0) = 1$$
$$\frac{dx_s}{dt} = \frac{u^+ + u^-}{2} = \frac{0 + \frac{x_s}{1+t}}{2} = \frac{x_s}{2+2t}$$

Solving the separable differential equation:

$$\frac{dx_s}{x_s} = \frac{dt}{2(1+t)}$$
$$\int \frac{1}{x_s}\, dx_s = \int \frac{1}{2(1+t)}\, dt$$
$$\ln|x_s| = \frac{1}{2}\ln|1+t| + C$$

Using the initial condition $x_s(0) = 1$:

$$\ln(1) = \frac{1}{2}\ln(1) + C$$
$$0 = 0 + C$$
$$C = 0$$

$$\ln|x_s| = \ln\left(\sqrt{1+t}\right)$$
$$x_s(t) = \sqrt{t+1}$$

Final piecewise solution:

$$u(x,t) = \begin{cases} 0, & x < 0 \\ \frac{x}{t+1}, & 0 \leq x < \sqrt{t+1} \\ 0, & x > \sqrt{t+1} \end{cases}$$

Question 3: Three Shocks

Enforcing the Rankine-Hugoniot condition, determine the shock solution of the IVP

$$\begin{cases} u_t + u^2 u_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}, \end{cases} \quad \text{where} \quad u_0(x) = \begin{cases} 2, & x < 0, \\ 1, & 0 \leq x \leq 1. \\ 0, & 1 < x. \end{cases}$$

Finding the flux function $\phi(u)$ by taking the integral:

$$u_t + \phi'(u)u_x = 0$$
$$\phi'(u) = u^2$$
$$\phi(u) = \int u^2 \, du = \frac{1}{3}u^3$$

For the first shock at $x_0 = 0$, taking the derivative of x_{s1}:

$$u^- = 2, \; u^+ = 1$$
$$\frac{dx_{s1}}{dt} = \frac{\phi(u^+) - \phi(u^-)}{u^+ - u^-} = \frac{\frac{1}{3}(1)^3 - \frac{1}{3}(2)^3}{1-2} = \frac{\frac{1}{3} - \frac{8}{3}}{-1} = \frac{7}{3}$$
$$dx_{s1} = \frac{7}{3}\,dt$$
$$\int dx_{s1} = \int \frac{7}{3}\,dt$$
$$x_{s1}(t) = \frac{7}{3}t + C$$

Using $x_{s1}(0) = 0$:

$$0 = \frac{7}{3}(0) + C \implies C = 0$$
$$x_{s1}(t) = \frac{7}{3}t$$

For the second shock at $x_0 = 1$, taking the derivative of x_{s2}:

$$u^- = 1, \; u^+ = 0$$
$$\frac{dx_{s2}}{dt} = \frac{\frac{1}{3}(0)^3 - \frac{1}{3}(1)^3}{0-1} = \frac{-\frac{1}{3}}{-1} = \frac{1}{3}$$
$$dx_{s2} = \frac{1}{3}\,dt$$
$$\int dx_{s2} = \int \frac{1}{3}\,dt$$
$$x_{s2}(t) = \frac{1}{3}t + C$$

Using $x_{s2}(0) = 1$:

$$1 = \frac{1}{3}(0) + C \implies C = 1$$
$$x_{s2}(t) = \frac{1}{3}t + 1$$

Solving for the collision time t_c:

$$\frac{7}{3}t = \frac{1}{3}t + 1$$
$$7t = t + 3$$
$$6t = 3$$
$$t_c = \frac{1}{2}$$
$$x_c = x_{s1}\left(\frac{1}{2}\right) = \frac{7}{6}$$

For the third shock, taking the derivative of x_{s3}:

$$u^- = 2,\ u^+ = 0$$
$$\frac{dx_{s3}}{dt} = \frac{\frac{1}{3}(0)^3 - \frac{1}{3}(2)^3}{0 - 2} = \frac{-\frac{8}{3}}{-2} = \frac{4}{3}$$
$$dx_{s3} = \frac{4}{3}\,dt$$
$$\int dx_{s3} = \int \frac{4}{3}\,dt$$
$$x_{s3}(t) = \frac{4}{3}t + C$$

Using the point $(x_c, t_c) = \left(\frac{7}{6}, \frac{1}{2}\right)$:

$$\frac{7}{6} = \frac{4}{3}\left(\frac{1}{2}\right) + C$$
$$\frac{7}{6} = \frac{4}{6} + C \implies C = \frac{3}{6} = \frac{1}{2}$$
$$x_{s3}(t) = \frac{4}{3}t + \frac{1}{2}$$

Final piecewise solution:

$$\text{For } t \leq \frac{1}{2}$$

$$u(x,t) = \begin{cases} 2, & x < \frac{7}{3}t \\ 1, & \frac{7}{3}t \leq x \leq \frac{1}{3}t + 1 \\ 0, & x > \frac{1}{3}t + 1 \end{cases}$$

$$\text{For } t > \frac{1}{2}$$

$$u(x,t) = \begin{cases} 2, & x < \frac{4}{3}t + \frac{1}{2} \\ 0, & x > \frac{4}{3}t + \frac{1}{2} \end{cases}$$

Question 4: Two shocks Enforcing the Rankine-Hugoniot condition, determine the shock solution of the IVP

$$\begin{cases} u_t + uu_x = 0, & x \in \mathbb{R}, t \geq 0, \\ u(x,0) = u_0(x), & x \in \mathbb{R}, \end{cases} \quad \text{where} \quad u_0(x) = \begin{cases} 1, & x < -1, \\ 0, & -1 \leq x \leq 1. \\ -1, & 1 < x. \end{cases}$$

$$u_t + uu_x = 0$$
$$\phi(u) = \frac{1}{2}u^2$$

For the first shock at $x_0 = -1$, taking the derivative of x_{s1}:

$$u^- = 1, \;\; u^+ = 0$$
$$\frac{dx_{s1}}{dt} = \frac{\frac{1}{2}(0)^2 - \frac{1}{2}(1)^2}{0-1} = \frac{-\frac{1}{2}}{-1} = \frac{1}{2}$$
$$dx_{s1} = \frac{1}{2}\,dt$$
$$\int dx_{s1} = \int \frac{1}{2}\,dt$$
$$x_{s1}(t) = \frac{1}{2}t + C$$

Using $x_{s1}(0) = -1$:

$$-1 = \frac{1}{2}(0) + C \implies C = -1$$
$$x_{s1}(t) = \frac{1}{2}t - 1$$

For the second shock at $x_0 = 1$, taking the derivative of x_{s2}:

$$u^- = 0, \;\; u^+ = -1$$
$$\frac{dx_{s2}}{dt} = \frac{\frac{1}{2}(-1)^2 - \frac{1}{2}(0)^2}{-1-0} = \frac{\frac{1}{2}}{-1} = -\frac{1}{2}$$
$$dx_{s2} = -\frac{1}{2}\,dt$$
$$\int dx_{s2} = \int -\frac{1}{2}\,dt$$
$$x_{s2}(t) = -\frac{1}{2}t + C$$

Using $x_{s2}(0) = 1$:

$$1 = -\frac{1}{2}(0) + C \implies C = 1$$
$$x_{s2}(t) = -\frac{1}{2}t + 1$$

Solving for the collision time t_c:

$$-\frac{1}{2}t + 1 = \frac{1}{2}t - 1$$
$$-t + 2 = t - 2$$
$$2t = 4$$
$$t_c = 2$$
$$x_c = x_{s2}(2) = -\frac{1}{2}(2) + 1 = 0$$

For the third shock, taking the derivative of x_{s3}:

$$u^- = 1,\ u^+ = -1$$
$$\frac{dx_{s3}}{dt} = \frac{\frac{1}{2}(-1)^2 - \frac{1}{2}(1)^2}{-1 - 1} = \frac{0}{-2} = 0$$
$$dx_{s3} = 0\,dt$$
$$\int dx_{s3} = \int 0\,dt$$
$$x_{s3}(t) = C$$

Using the point $(x_c, t_c) = (0, 2)$:

$$C = x_c = 0$$
$$x_{s3}(t) = 0$$

Final piecewise solution:

$$\text{For } t \leq 2$$
$$u(x,t) = \begin{cases} 1, & x < \frac{1}{2}t - 1 \\ 0, & \frac{1}{2}t - 1 \leq x \leq -\frac{1}{2}t + 1 \\ -1, & x > -\frac{1}{2}t + 1 \end{cases}$$

$$\text{For } t > 2$$
$$u(x,t) = \begin{cases} 1, & x < 0 \\ -1, & x > 0 \end{cases}$$

Question 5: Feedback

How long did you spend on this homework? Was it easy/medium/hard? Do you have any feedback for the instructor/TA? Do you like the inclusion of bonus problems?

This homework took me a little while but that was because I was busy with work from other classes. I liked how the problems on this homework were mostly the same style. I have no opinion on the bonus questions since I normally don't have time to do them.